

ACQ580

Variable Speed Drive for Water & Wastewater

Complete Technical & Integration Reference Guide

Covers: Operating Principles · Physical Setup · Modbus RTU Configuration · Register Mapping · Transducer Wiring · IoT Gateway Integration · EMI Protection · RS485 Best Practices · Pump Station Bill of Materials

Document type	Engineering Reference
Drive model	ABB ACQ580-01 (all frame sizes)
Communication	Modbus RTU / RS485 (embedded EFB)
Application	Water & Wastewater Pump Stations

Table of Contents

1	How the ACQ580 Works
2	First-Time Setup & Commissioning
3	Modbus RTU Configuration
4	Modbus Parameters & Register Map
5	Reading Data via USB–RS485 on a PC
6	Connecting a Pressure Transducer
7	IoT Gateway Connection
8	RS485 Ground (DGND) — Mandatory or Optional?
9	EMI from the ACQ580 — Effects & Protection
10	Non-Opto-Isolated RS485 — Risks & Precautions
11	Gateway Distance & Safe Placement
12	Complete Pump Station Physical Setup
13	Bill of Materials — Surrounding Equipment List

Section 1

How the ACQ580 Works

The ACQ580 is a Variable Speed Drive (VSD) — also called a Variable Frequency Drive (VFD) — designed specifically for water and wastewater pump applications. It converts fixed-frequency mains AC power into variable-frequency, variable-voltage AC output to control motor speed and torque precisely, enabling significant energy savings and advanced pump protection.

Internal Power Conversion Stages

Stage	What Happens	Key Component
1. Rectifier	Converts 3-phase AC mains (230–480 V) to DC via diode bridge rectifier	4× 125 J MOVs for surge protection
2. DC Bus	Capacitor bank stores energy; provides stable DC link (~1.35 × mains V)	Electrolytic capacitors
3. IGBT Inverter	Switches DC at 4–12 kHz using PWM to create variable-frequency 3-phase AC (0–50 Hz)	IGBTs (40–50 A)
4. Control Board	Runs pump control firmware; handles PID, I/O, fieldbus, all protections	Embedded microcontroller

Pump-Specific Features Built In

- **PID loop (Process control):** Takes 4–20 mA feedback from pressure/flow transducer on AI2; automatically adjusts motor speed to maintain setpoint.
- **Anti-cavitation protection:** Detects cavitation and reacts by reducing speed.
- **Dry-run protection:** Stops the pump if water is absent — via DI signal or current monitoring.
- **Soft pipe fill:** Ramps up slowly on start to prevent water hammer.
- **Multi-pump (IPC):** Controls up to 3 pumps in rotation with drive-to-drive D2D communication.
- **Sensorless flow calculation:** Estimates flow from motor parameters — no flow meter needed.
- **Safe Torque Off (STO):** Hardware safety function built in as standard.
- **EMC filter:** Built-in EMC filter (EN 61800-3 compliant, enabled by default for TN-S supplies).

Operating Modes

Mode	Controlled By	Typical Use
Local (Hand)	ACX-AP control panel on drive	Commissioning, maintenance, manual override
Remote (Auto / EXT1)	I/O terminals or Modbus RTU fieldbus	Normal automated pump station operation
Remote (Auto / EXT2)	Second external source (e.g. fieldbus override)	Backup or process override scenarios

Section 2

First-Time Setup & Commissioning

■ **Warning:** Only qualified electrical personnel should install and commission this drive. Dangerous voltages remain present for up to 5 minutes after power-off. Always verify DC bus voltage is zero with a multimeter before working inside.

Step-by-Step Commissioning Sequence

1. Mechanical mounting

Mount vertically on a flat, vibration-free surface. Minimum clearances: 100 mm top & bottom, 50 mm sides. Operating temperature: –15 to +50 °C. Do not restrict airflow over the heatsink.

2. Power wiring (mains input)

Connect 3-phase supply (L1, L2, L3) to input terminals U1, V1, W1. Connect PE to the drive's PE terminal. Use copper conductors sized per the hardware manual fuse tables.

3. Motor wiring (output)

Connect motor to output terminals U2, V2, W2. Always use shielded VFD-rated motor cable. Connect cable shield to PE at both drive and motor ends. Keep motor cable as short as possible.

4. Control wiring

Connect I/O signals, pressure transducer (4–20 mA to AI2), and RS485 fieldbus to terminal block X2.

5. Primary Settings wizard

On first power-up the ACX-AP panel starts the wizard. Set language, application type (select 'Water' default), and enter motor nameplate data (voltage, current, frequency, speed, power).

6. Motor ID run

Perform ID run via Menu → Primary settings → ID run. The drive measures motor parameters automatically. Use rotating ID run if possible for best results.

7. Set control parameters

Set control source (EXT1 / Modbus), reference source, PID setpoint source and scaling, and verify all protection thresholds (overload, dry-run, cavitation limits).

8. Test run

Start in Hand (local) mode. Verify rotation direction. Confirm smooth ramp-up/ramp-down. Check that PID control maintains setpoint before switching to Auto mode.

Control Terminal Block X2 — Key Pins

Terminal	Label	Function
X2:2 / X2:3	AI1+ / AI1–	Analogue input 1 (0–10 V or 0/4–20 mA) — speed setpoint
X2:4 / X2:5	AI2+ / AI2–	Analogue input 2 — pressure transducer feedback (4–20 mA)
X2:6 / X2:7	AO1 / AGND	Analogue output (0–20 mA) — e.g. output frequency signal
X2:10	+24V	Auxiliary 24 VDC output, max 250 mA — for transducer supply
X2:11	DGND	Digital ground

Terminal	Label	Function
X2:14	DI1	Digital input 1 — default: Start / Stop command
X2:15	DI2	Digital input 2 — default: Run permissive / dry-run contact
X2:24	B+	RS485 Modbus B+ (non-inverting / Data+)
X2:25	A–	RS485 Modbus A– (inverting / Data–)
X2:26	DGND	RS485 signal ground reference — connect to master/gateway GND
X2:27	TERM	Termination switch — set ON if this drive is at an end of the bus

Section 3

Modbus RTU Configuration

■ **Note:** The ACQ580 has a built-in Embedded Fieldbus (EFB) interface supporting Modbus RTU as standard. No additional adapter card is needed. Physical interface: EIA-485 (RS485), half-duplex, 2-wire + signal ground. For advanced protocols (PROFIBUS, EtherNet/IP) the optional FSCA-01 adapter can be used.

RS485 Wiring

Use Belden 9842 or equivalent: dual twisted shielded pair, 120 Ω nominal impedance. Connect one twisted pair to B+ (X2:24) and A– (X2:25). Use the second pair or a separate wire for DGND (X2:26).

Drive Terminal	Connect to Master / Gateway	Notes
X2:24 — B+	RS485 B+ (Data+, non-inverting)	Twisted pair — wire 1
X2:25 — A–	RS485 A– (Data–, inverting)	Twisted pair — wire 2
X2:26 — DGND	Signal GND at master/gateway	3rd wire or 2nd pair — see Section 8
Termination switch	SW = ON if drive is at bus end	Built-in 120 Ω termination resistor

■ **Warning:** B+/A– naming confusion: ABB labels B+ as non-inverting (logic high when idle) and A– as inverting. Some gateways label their terminals the opposite way. Always verify polarity with a multimeter before first power-on of the bus. Reversed polarity causes no hardware damage but communication will fail.

Parameter Group 58 — Embedded Fieldbus Settings

Parameter	Name	Setting for Modbus RTU	Notes
58.01	Protocol	Modbus RTU	Selects Modbus RTU as EFB protocol
58.03	Node address	1–247 (unique per bus)	Must match master's slave address setting
58.04	Baud rate	9600 / 19200 / 38400	Default: 19200. Must match master.
58.05	Parity / stop bits	8 Even 1 (default)	Must match master. 8E1 is Modbus standard.
58.06	Refresh settings	Set to 'Refresh' after any change	CRITICAL: changes don't apply until refreshed or power-cycled.
58.14	Comm loss timeout	3–10 s recommended	Time before drive reacts to communication loss
58.15	Comm loss function	Fault / Last speed / Stop	Use 'Last speed' for safety in pumping applications
58.33	Addressing mode	0 (default)	Allows both 16-bit and 32-bit register access simultaneously

Enable Modbus Control (not just monitoring)

Parameter	Set To	Purpose
20.01	EFB	Command source = fieldbus. Allows start/stop via Modbus.
28.11	EFB Ref1	Frequency reference from fieldbus. Allows speed control via Modbus.
19.11	EXT1	Selects EXT1 as active control location.

■ **Warning:** If P20.01 is not set to EFB, the drive will acknowledge Modbus commands but will NOT start. If P28.11 is not set to EFB Ref1, writing a speed/frequency value over Modbus will have no effect. These are the two most common

Modbus commissioning mistakes.

Section 4

Modbus Parameters & Register Map

Register Address Formula

Type	Formula	Example
16-bit read/write	$40000 + (\text{group} \times 100) + \text{index}$	P01.06 → $40000 + (1 \times 100) + 6 = 40106$
32-bit read/write (P58.33=0)	$420000 + (200 \times \text{group}) + (2 \times \text{index})$	P01.06 → $420000 + (200 \times 1) + (2 \times 6) = 420212$

Essential Read-Only Monitoring Registers (16-bit)

Modbus Reg	Parameter	Name	Unit / Scaling	Conversion
40101	01.01	Motor speed used	rpm, 0–20000	$\div 20000 \times P46.01$ (rated speed)
40106	01.06	Output frequency	Hz, 0–20000	$\div 20000 \times P46.02$ (rated freq)
40107	01.07	Motor current	A, see note	Set P46.05=100 → $\div 100$ for 0.01 A resolution
40110	01.10	Motor torque	%, 0.01	$\div 100 = \%$ torque
40113	01.13	Output voltage	V, 1 V	No conversion
40114	01.14	Output power	kW, 0.1	$\div 10 = \text{kW}$
40111	01.11	DC bus voltage	VDC, 0.1	$\div 10 = \text{VDC}$
40120	01.20	Inverter kWh counter	kWh, 0.1	$\div 10 = \text{kWh}$
40503	05.03	Hours run	hours	Whole numbers, no conversion
40511	05.11	Inverter temperature	%, 1%	100% = overtemperature fault
40616	06.16	Drive status word 1	Bit flags	Bit 8=Local, Bit 10=Ext1, Bit 11=Ext2
40622	06.22	Hand-Off-Auto status	Bit flags	Bit 0=Hand, Bit 1=Off, Bit 2=Auto
40411	04.11	Last fault code	Fault code	Convert to hex, compare to fault list in firmware manual
40412	04.12	2nd to last fault	Fault code	Convert to hex
40814	08.14	PID actual value (feedback)	Process units	Scaled per AI2 configuration
40815	08.15	PID setpoint	Process units	Scaled per setpoint source configuration

Control Registers (Read / Write)

Modbus Reg	Name	Function
40001	Control Word	Start/stop/reset: Bit 0=On/Off, Bit 3=Enable, Bit 7=Fault reset
40002	Speed Reference 1	Speed/frequency setpoint. 0–20000 maps to 0–rated speed/frequency.
40003	Speed Reference 2	Alternative reference 2.

Modbus Function Codes Supported

Code	Name	Use
03 (0x03)	Read Holding Registers	Read parameters and status — most common
04 (0x04)	Read Input Registers	Read analogue input values
06 (0x06)	Write Single Register	Write one parameter or control register
16 (0x10)	Write Multiple Registers	Write control word + reference in a single frame
01 (0x01)	Read Coils	Read individual control word bits
05 (0x05)	Write Single Coil	Write individual control word bits

Section 5

Reading Data via USB–RS485 on a PC

■ **Note:** Yes — a USB to RS485 converter connected to the drive's EFB port (B+, A–, DGND) lets you read and write parameters, monitor the drive, and even control it — without a PLC, gateway, or ABB Drive Composer software.

What You Need

- **USB to RS485 converter:** Use a converter with a genuine FTDI FT232 chip, or an isolated version (e.g. FTDI USB-RS485-WE, Waveshare isolated USB-RS485, or equivalent).
- **Software:** Any Modbus RTU master tool — ModbusPoll (Windows), QModMaster (cross-platform), modscan32, or a Python script using the pymodbus library.
- **Wiring:** Converter A(–) → drive X2:25, Converter B(+) → drive X2:24, Converter GND → drive X2:26.

PC Software / Serial Port Settings

Setting	Value
Port	COM port assigned to USB-RS485 adapter (check Device Manager on Windows)
Protocol	Modbus RTU
Baud rate	Match drive P58.04 (default: 19200)
Data bits	8
Parity	Even (match P58.05, default: 8E1)
Stop bits	1
Slave ID	Match drive P58.03 (default: 1)
Response timeout	500–1000 ms

Python Example — Read Output Frequency

```
from pymodbus.client import ModbusSerialClient

client = ModbusSerialClient(
    port='COM3', baudrate=19200, parity='E',
    stopbits=1, bytesize=8, timeout=1)
client.connect()

# Register 40106 → 0-based address = 105
result = client.read_holding_registers(address=105, count=1, slave=1)
freq_raw = result.registers[0]
freq_hz = freq_raw / 20000 * 50 # rated freq = 50 Hz (P46.02)
print(f'Output frequency: {freq_hz:.2f} Hz')
client.close()
```

■ **Warning:** Use a fully isolated USB-RS485 converter. A non-isolated converter references your laptop ground to the drive's DGND (which is near the high-voltage section) — this creates a shock hazard and introduces noise. Isolated converters cost only slightly more and are significantly safer.

Section 6

Connecting a Pressure Transducer

The most common sensor in a pump station is a 4–20 mA pressure transducer connected to AI2 on the drive for PID process feedback — enabling the drive to maintain constant discharge pressure.

Wiring — 3-wire, 4–20 mA Loop-Powered Transducer

Transducer Wire	Connect to Drive	Notes
+ (Signal / Power)	X2:4 (AI2+)	4–20 mA signal into the drive current input
– (Return / GND)	X2:5 (AI2–) and X2:7 (AGND)	Signal return and analogue ground
Shield	X2:1 (SCR)	Connect shield at drive end only; leave far end floating
Supply + (if 3-wire)	X2:10 (+24V)	Drive auxiliary 24 VDC output (max 250 mA)
Supply – (if 3-wire)	X2:11 (DGND)	Drive digital ground as supply return

Drive Parameters for AI2 Pressure Feedback & PID

Parameter	Set To	Purpose
12.12	4–20 mA	Configures AI2 as a 4–20 mA current input
12.13	Min value (e.g. 0.0 bar)	Corresponds to 4 mA signal from transducer
12.14	Max value (e.g. 10.0 bar)	Corresponds to 20 mA signal from transducer
40.07 (PID feedback src)	AI2	Sets AI2 as the PID process feedback source
40.08 (PID setpoint)	e.g. 6.0 bar	Target pressure to maintain

■ **Note:** Transducer selection: Use a 4–20 mA output transducer rated for the maximum system pressure × 1.2 safety margin. For a pump rated to 10 bar, use a 0–16 bar transducer. Ensure IP67 or better for outdoor or wet pump station environments.

Section 7

IoT Gateway Connection via Modbus RTU

An IoT or Modbus gateway (e.g. Advantech ADAM, Moxa NPort/MGate, Ewon Cosy, Weintek cMT, Robustel R3000, or a custom RS485→MQTT gateway) acts as the Modbus master. The ACQ580 is the Modbus slave.

Connection Topology

Gateway Port	Drive Terminal	Signal
RS485 B+ (Data+)	X2:24 — B+	Twisted pair, wire 1
RS485 A- (Data-)	X2:25 — A-	Twisted pair, wire 2
GND / Signal GND	X2:26 — DGND	3rd wire or second pair
120 Ω termination	Enable at gateway if it is the bus end	Required at both physical ends of the RS485 bus

Gateway Configuration Settings

- **Protocol:** Modbus RTU (not ASCII, not Modbus TCP)
- **Baud rate:** Match drive P58.04 — recommend 9600 for noisy environments, 19200 default
- **Parity:** Match drive P58.05 (default 8E1 — 8 data, even parity, 1 stop bit)
- **Slave address:** Match drive P58.03 (default: 1)
- **Poll interval:** Minimum 100 ms; 500 ms–1 s is typical for pump monitoring
- **Response timeout:** 500–1000 ms
- **Register type:** Holding Registers (Function Code 03) for all ACQ580 parameters
- **Register addressing:** 0-based — subtract 40001 from the Modbus register number (e.g. register 40106 → enter address 105 in the gateway)

■ **Warning:** 0-based vs 1-based addressing: Some gateways use 1-based addressing where '40106' is entered as-is. Others use 0-based where you enter 105. Check your gateway manual carefully. An incorrect offset is the #1 cause of 'register not found' errors.

Section 8

RS485 Ground (DGND) — Mandatory or Optional?

■ **Warning:** DGND is mandatory for reliable communication in industrial environments with drives, motors, and switching equipment. Omitting it is the second most common cause of RS485 communication problems.

Why DGND Matters

RS485 is a differential signalling standard — it can tolerate common-mode noise up to ± 7 V. However, without a common ground reference between master and slave, ground potential differences can easily exceed this limit near VSDs (which generate significant common-mode voltage). This results in corrupted data, CRC errors, and intermittent communication failures.

Correct DGND Connection Rules (per ABB ACQ580 Manual)

1. Connect a DGND conductor from drive X2:26 to the signal ground reference at the automation controller or gateway.
2. Use a single DGND connection per device — do not create ground loops by connecting DGND at multiple points.
3. Connect cable shields at each drive via the SCR terminal (X2:1). Do NOT connect the shield to the drive's PE/chassis ground.
4. Connect the combined cable shield to automation controller/gateway ground at one end only.
5. Do NOT ground the RS485 network directly to earth at more than one point along the bus.

Termination Resistors (120 Ω)

- Required at **both physical ends** of the RS485 bus. Drives at mid-bus: termination OFF.
- The ACQ580 has a built-in 120 Ω termination switch at terminal X2:27 — set to ON for end-of-bus drives.
- Only **one device** (typically the master/gateway) should enable bias resistors (pull-up/pull-down) to define the idle bus state.
- Maximum recommended bus length: 1200 m at 9600 baud; 300 m at 38400 baud.

Section 9

EMI from the ACQ580 — Effects & Protection

■ **Warning:** The ACQ580 is a significant EMI source. IGBT switching at 4–12 kHz creates high-frequency electromagnetic interference that can corrupt RS485 communication, damage gateway circuits, and cause false readings in pressure transducers.

EMI Sources in the ACQ580

- **IGBT switching transients:** Fast dV/dt on the motor cable creates conducted and radiated EMI at harmonics of the switching frequency (4–12 kHz and beyond).
- **Common-mode current:** Flows through motor cable capacitance to PE, generating high-frequency ground currents that contaminate signal cables sharing the same conduit or tray.
- **Motor cable radiation:** Unshielded or poorly shielded motor cable acts as an antenna for radiating switching noise.
- **DC bus switching noise:** Visible on supply lines as conducted emissions — can affect other equipment on the same bus.

■ **Note:** The ACQ580 complies with EN 61800-3 (EMC product standard for drives). However, EMC compliance only holds under correct installation conditions — proper shielding, cable routing, and grounding as described below.

How EMI Affects the Gateway and RS485 Bus

- Induced noise on RS485 cables → CRC errors, data corruption, communication dropouts
- Common-mode voltage transients → can permanently damage non-isolated RS485 interface chips
- High-frequency ground loop current → corrupts the signal reference voltage
- Physical proximity to motor cables → direct capacitive and inductive coupling

EMI Protection Measures

Measure	How to Apply	Priority
Shielded RS485 cable	Use Belden 9842; connect shield at one end only	Critical
Cable separation	RS485 cable in separate tray from motor cable; minimum 300 mm	Critical
Avoid parallel runs	Never run RS485 parallel to motor cable for more than 1 m	Critical
Shielded motor cable	Always use shielded VFD-rated motor cable; shield to PE at both ends	Critical
90° crossings	Where signal and power cables must cross, cross at right angles	Important
Ferrite cores	Snap-on ferrite cores on RS485 cable near the drive and at the gateway	Important
Drive EMC filter	Ensure drive EMC filter is not disabled (factory default: enabled for T115)	Important
Panel grounding	Star-point grounding — all PE/shield connections to a single ground bar	Important
Isolated RS485 gateway	Use gateway with optically isolated RS485 port (see Section 10)	Strongly recommended
RS485 SPD	Install ABB OVR RS485 or equivalent surge protective device at gateway	Critical for non-isolated

Section 10

Non-Opto-Isolated RS485 in Gateway — Risks & Precautions

Risks of Non-Isolated RS485 Near a VSD

Risk	Description
Transceiver chip damage	VSD-generated common-mode transients can exceed the RS485 chip's ± 7 V tolerance, permanently destroying the transceiver.
Ground loop currents	Current flowing through the ground conductor between drive and gateway adds noise to data signals, causing CRC errors.
Data corruption	High-frequency switching noise couples into the non-isolated RS485 line, corrupting Modbus frames.
Gateway instability / lockup	Severe ground disturbances can freeze the gateway processor, requiring a power cycle to recover.

Pros of Non-Isolated RS485

- Lower gateway cost
- Simpler implementation — no isolation barrier to manage
- Acceptable in low-noise environments far from large drives (e.g. office automation, small PLC panels)

Precautions — Non-Isolated RS485 Near ACQ580

#	Precaution	Why
1	Install an RS485 SPD (Surge Protective Device) at the gateway RS485 port.	Absorbs VSD surge energy (10 kA, 160 V) to protect the non-isolated transceiver.
2	Add a signal-line common-mode choke (ferrite core) on the RS485 cable.	Reduces high-frequency conducted noise.
3	Keep DGND as a single conductor — do not create ground loops.	Prevents circulating ground currents on the bus.
4	Route RS485 cable well away from motor cable and drive (see Section 10.2.1 for distances).	Reduces capacitive and inductive coupling.
5	Use shielded Belden 9842 cable; connect shield at gateway end to gateway chassis ground.	Shields against radiated EMI.
6	Power the gateway from a separate, clean 24 VDC SMPS — not from the drive's power line.	Prevents power-line noise coupling into the gateway CPU.
7	If problems persist after all precautions, upgrade to a gateway with optically isolated RS485.	The isolated RS485 is a long-term solution.

Section 11

Gateway Distance & Safe Placement from the VSD

Minimum Recommended Distances

Component / Scenario	Minimum Distance	Reason
IoT Gateway (non-isolated RS485) — physical separation from drive enclosure	1 m	Radiated EMI from drive heatsink and capacitor bank
IoT Gateway (isolated RS485) — in same panel	Separate compartment	Cable segregation removes most risk; wiring segregation still required
RS485 cable parallel run alongside motor cable	≥ 300 mm separation	Inductive coupling from high motor cable current harmonics
RS485 cable from drive output terminals	≥ 200 mm	High dV/dt switching noise radiation near output busbars
Gateway from motor cable conduit	≥ 300 mm	Magnetic field from motor cable current harmonics

Physical Mounting Recommendations

- Mount the gateway in the **upper section** of the panel or in a **separate enclosure** from the VSD.
- Use a **metal DIN rail enclosure** for the gateway — metal provides electromagnetic shielding.
- Connect the gateway enclosure to the PE/chassis ground star point.
- If drive and gateway must share one panel: install a **grounded metal divider plate** between the VSD section and the gateway/controls section.
- Route all signal cables (RS485, 4–20 mA, DI/DO) in a separate metallic conduit or cable tray from motor power cables.

Section 12

Complete Pump Station Physical Setup

The following describes a complete pump station installation using the ACQ580 VSD, a pressure transducer for PID feedback, and an IoT Modbus gateway for cloud monitoring and control.

Panel Layout (from top to bottom, or left to right)

Zone	Equipment	Notes
Power entry zone	3-phase MCB/fuses, input line reactor (optional)	All mains-rated components; rated copper busbars
Drive zone	ACQ580 VSD (main enclosure)	Minimum clearances as per manual; heatsink ventilation unobstructed
Divider plate	Grounded metal plate	Separates HV power zone from LV signal/control zone
Signal/control zone	24 VDC SMPS, terminal blocks, I/O wiring	All 24 VDC signals; separate from power zone
Gateway zone	IoT Modbus gateway, RS485 SPD on DIN rail	If non-isolated RS485: install SPD immediately at gateway RS485 port
Ground bar	PE/ground busbar (copper)	Star-point grounding for all shields, PE conductors, VSD PE

Signal & Cable Routing Rules

- **Motor cable (U2, V2, W2 + PE):** Shielded VFD-rated cable; shield grounded at PE star bar at both drive and motor ends; run in its own metallic conduit, minimum 300 mm from signal cables.
- **Mains input (U1, V1, W1):** Standard power cable; separate conduit from motor cable.
- **4–20 mA transducer cable:** Shielded; separate from motor cable; shield grounded at drive end (X2:1) only; leave far end floating.
- **RS485 Modbus cable:** Belden 9842 shielded; completely separate conduit from motor cable; shield connected at one end only; install SPD at gateway entry.
- **Gateway power supply cable:** From a clean 24 VDC SMPS; NOT from drive auxiliary +24V output.

Wiring Segregation Summary

Cable Type	Conduit / Tray	Min. Separation from Motor Cable
Motor power cable	Conduit 1 (metal)	N/A — reference
Mains input cable	Conduit 2 (metal)	Preferred ≥ 150 mm
4–20 mA signal cable	Conduit 3 (metal)	≥ 300 mm
RS485 Modbus cable	Conduit 3 (metal)	≥ 300 mm
24 VDC control wiring	Conduit 3 (metal)	≥ 300 mm

Section 13

Bill of Materials — Surrounding Equipment List

Item	Specification	Notes
ABB ACQ580-01 VSD	Frame size matched to motor kW and mains voltage	Includes built-in EMC filter, STO, EFB Modbus RTU — no extra cabling
3-phase MCB or fuses	GG fuses or type B/C MCB; rated per ACQ580 hardware input protection	ABB provides fuse/MCB tables by drive motor rating
Input contactor (optional)	AC-3 rated; sized per drive input current rating	Do NOT use for frequent start/stop — use drive start command; consider VFD series
Shielded motor cable	3-phase + PE; VFD-rated low-capacitance cable, e.g. Belden 9842 or equivalent	Keeps VFD noise in; use output dV/dt choke for cables > 50 m
Output dV/dt choke (optional)	ABB output filter; rated for drive output current	Reduces motor winding stress from switching transients; required for cables > 50 m
Input line reactor (optional)	3–5% impedance; rated for drive input current	Reduces harmonic distortion on supply; protects against supply voltage dips
Pressure transducer	4–20 mA, 0–16 bar (or as required), IP67 minimum, 0–24 VDC supply	Connect to drive X2; power from drive +24V (X2:10VDC); signal to drive X2:20mA
RS485 cable	Belden 9842 or equivalent: dual twisted pair, shielded, 120 Ohm characteristic impedance	Run from drive to gateway; critical for reliable communication
RS485 SPD (Surge Protective Device)	ABB OVR RS485 (10 kA, IEC 61643-21) or equivalent	Install at gateway RS485 port; absolutely essential for non-isolated RS485
IoT / Modbus gateway	RS485 Modbus RTU master with Ethernet/WiFi/4G; configurable, optoisolated RS485 port	Address must match P58.03; ensure isolation between VSD and gateway
24 VDC SMPS for gateway	DIN rail SMPS, 24 V output, 1–2 A minimum	Separate from drive auxiliary output — prevents power-line noise
DIN rail enclosure / panel	Metal enclosure IP54 or better; for outdoor/wet pump stations	Contains 24VDC SMPS and RS485 SPD; use IP65-rated panel
PE / Ground busbar	Copper busbar with suitable lugs; properly bonded to panel chassis	Grounding for all PE conductors, cable shields, and VSD frame
Cable conduit / tray	Separate metallic conduit for: (1) power/motor, (2) signal/RS485, (3) compressed air	Prevents signal interference; maintain separation between power and signal conduits
Ferrite core chokes	Snap-on ferrite; Würth WE-CBH or TDK equivalent; sized for RS485 cable type and length	Install on RS485 cable at drive entry and at gateway entry
USB to RS485 adapter (PC access)	Isolated USB-RS485; FTDI chip recommended (e.g. Wavecomm isolating and ESD-protected)	Use only if required for configuration; use isolated adapter
ACX-AP-W control panel (optional)	Remote panel with IP54 cover for external wall mounting	For local control outside the main enclosure; connects via panel bus
Motor thermistor protection	PTC or NTC thermistor embedded in motor windings	Provides thermal protection to the motor; required for motors in submerged or remote locations
Motor circuit protector (optional)	Motor protection relay or electronic overload relay	Provides additional overcurrent protection independent of drive; required for certain motor types

Official ABB Documentation References

Document	ABB Code	Where to Find
ACQ580-01 Hardware Manual	3AXD50000044862	library.e.abb.com — search document code
ACQ580 Pump Control Firmware Manual (Modbus compatible)	3AXD50000035867	library.e.abb.com — search document code
ACQ580-01 Quick Installation Guide (R1–R5)	3AXD50000044864	library.e.abb.com — search document code
Surge Protection Technical Note (ACQ580)	Technical Note 080	library.e.abb.com — search 'Technical Note 080'
Modbus Register Conversion 550→580	Technical Note 062	library.e.abb.com — search 'Technical Note 062'
ACQ580 Product Page	—	www.abb.com → Motion → Drives → Water → ACQ580